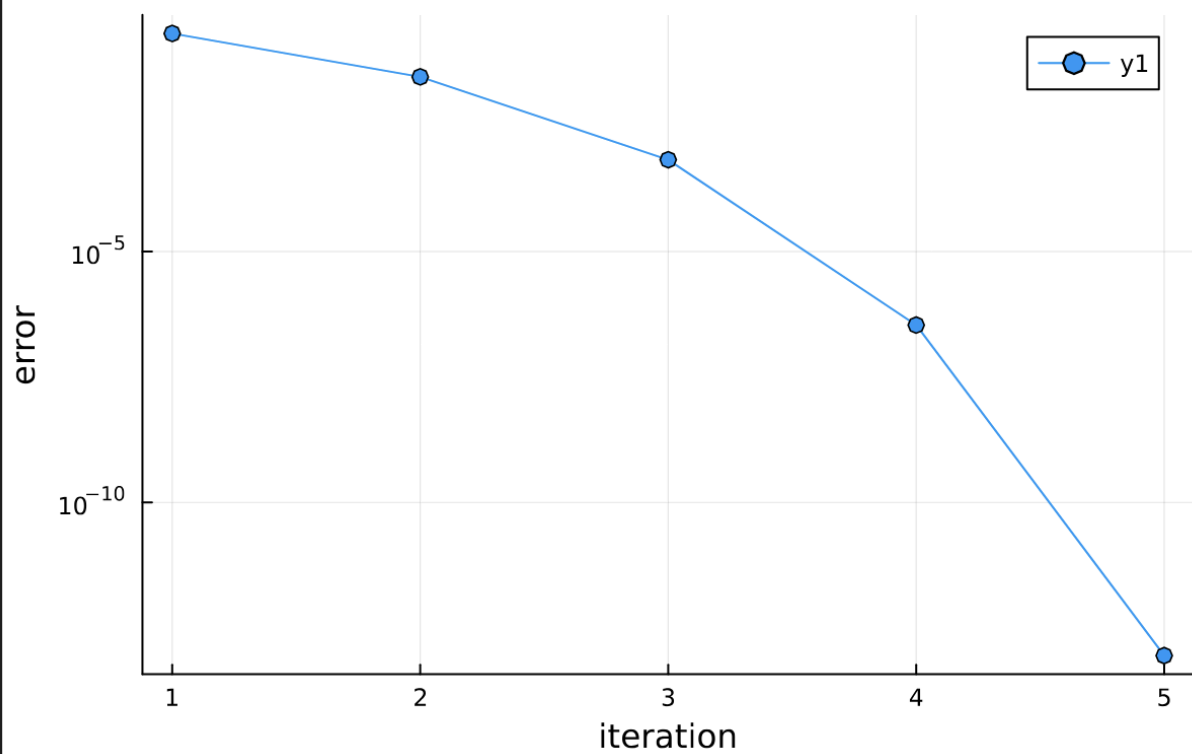


Quadratic Convergence for Newton's Method ($\gamma = 4.0$)



Note that this is just Newton's in multiple dimensions, and we want to set both of these equations to 0. The first equation we set to zero is obviously $x = \gamma x(1 - \tanh(x))$, which setting to zero (we know from above) is $1 = \gamma(1 - \tanh(x))$. So, the first equation is $f_1(x, \gamma) = \gamma(1 - \tanh(x)) - 1 = 0$

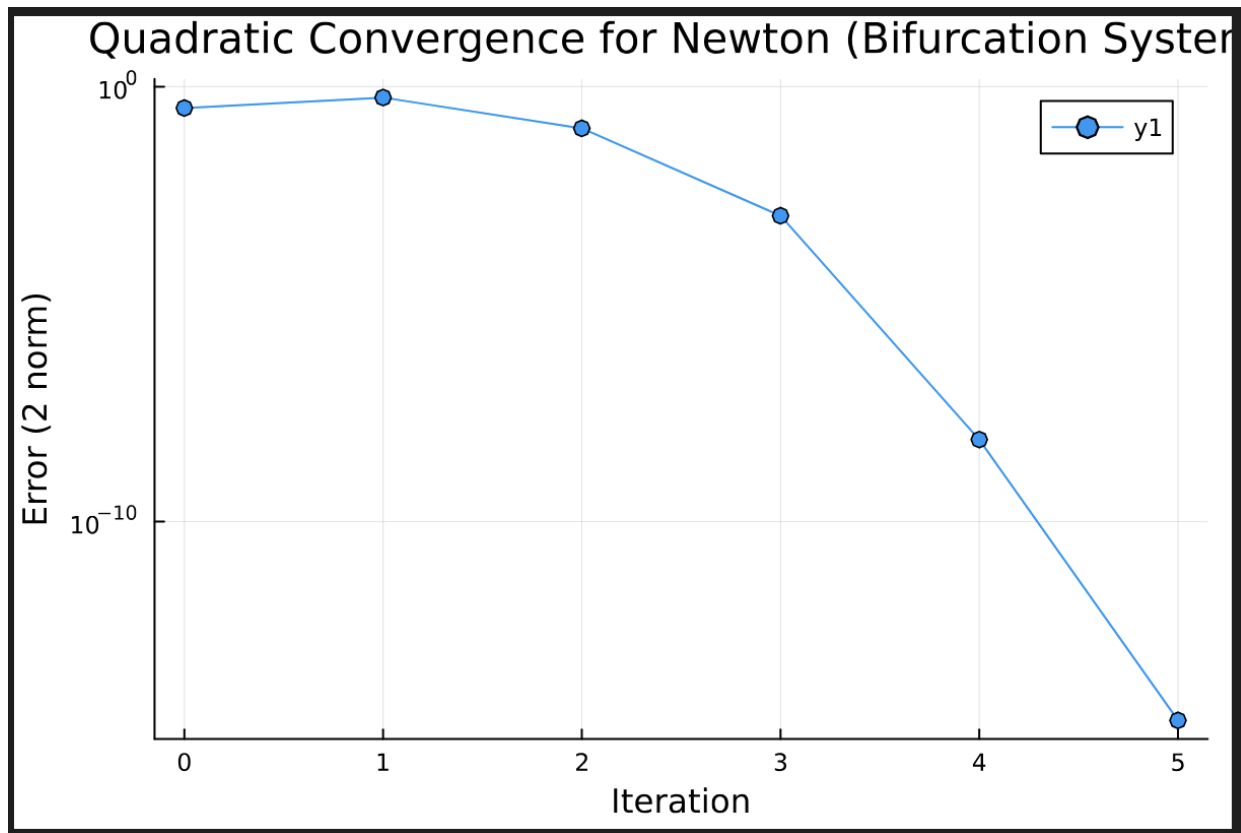
The second equation, we have to first compute the derivative, which we know will be with respect to x and a fixed parameter γ . Using the product rule, we will get that $G'(x; \gamma) = \gamma(1 - \tanh(x)) - \gamma x \operatorname{sech}(x)^2$, which knowing that $1 = \gamma(1 - \tanh(x))$, simplify so that it is $G'(x; \gamma) = 1 - \gamma x \operatorname{sech}(x)^2$. Set this equal to -1 to get the second equation, $f_2(x, \gamma) = \gamma x \operatorname{sech}(x)^2 - 2 = 0$

If we collect the two functions into a vector-valued-function, we will get that $f(x, \gamma) = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$

For the Newton's $(x_{k+1} = x_k - J(f(x, \gamma))^{-1} f(x, \gamma))$, the Jacobian is

$$J = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial \gamma} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial \gamma} \end{pmatrix} = \begin{pmatrix} -\gamma \operatorname{sech}^2(x) & 1 - \tanh(x) \\ \gamma \operatorname{sech}^2(x)(1 - 2x \tanh(x)) & x \operatorname{sech}^2(x) \end{pmatrix}$$

It begins that oscillations around 5, so choose that to be initial guess, although we can choose any close/good guess if we want (and use the previous guess for the initial x)



If extra justification is needed, note that we know that the function is already smooth (obviously, which is why we can use this method), but when approximating, the Jacobian isn't close to singular at the solution, as the determinant of the Jacobian isn't even close to 0

$$G(x \pm z) \approx G(x) \pm G'(x)z + \frac{1}{2}G''(x)z^2$$

Since $G(x) = x$, we can substitute that in to get:

$$G(x \pm z) \approx x \pm G'(x)z + \frac{1}{2}G''(x)z^2$$

Substitute the conditions in (1 will suffice), which says that $u = G(v)$ (approximately as the problem says), which is the same thing as saying:

$$x + z = G(x - z) \approx x - G'(x)z + \frac{1}{2}G''(x)z^2$$

Canceling out both sides, we will get $z(1 + G'(x)) = \frac{1}{2}G''(x)z^2$, which means the final equation is:

$$z \approx \frac{2(1 + G'(x))}{G''(x)}$$

Note that for the Newton's Method, we want to do the multivariate Newton's Method as detailed meticulously above, but with:

$$F(u, v) = \begin{pmatrix} u - G(v) \\ v - G(u) \end{pmatrix}$$

to set equal to the 0 vector